

Patterns of natural ^{15}N in soils and plants from chemically and organically fertilized uplands

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Abstract

Diagnostic tests for organic production of crops would be useful. In this study, the difference in natural ^{15}N abundances ($\delta^{15}\text{N}$) of soils and plants between fertilizer-applied upland (FU) and compost-applied upland (CU) fields was investigated to study using $\delta^{15}\text{N}$ as a marker of organic produce. Twenty samples each of soils and plants were collected from each field in early summer after applying fertilizer or compost. The $\delta^{15}\text{N}$ of fertilizers and composts was $-1.6 \pm 1.5\text{‰}$ ($n = 8$) and $17.4 \pm 1.2\text{‰}$ ($n = 10$), respectively. The $\delta^{15}\text{N}$ of total soil-N was significantly ($P < 0.05$) higher in CU fields ($8.8 \pm 2.0\text{‰}$) than in FU fields ($5.9 \pm 0.7\text{‰}$) due to long-term continuous application of ^{15}N -enriched compost, as indicated by a positive correlation ($r = 0.62$) between N content and $\delta^{15}\text{N}$ of total soil-N. The NO_3^- pool of CU soils ($11.6 \pm 4.5\text{‰}$) was also significantly ($P < 0.05$) enriched in ^{15}N compared to FU soils ($4.7 \pm 1.1\text{‰}$), while the ^{15}N contents of NH_4^+ pool were not different between both soils. Compost application resulted in ^{15}N enrichment of plants; the $\delta^{15}\text{N}$ values were $14.6 \pm 3.3\text{‰}$ for CU and $4.1 \pm 1.7\text{‰}$ for FU fields. These results showed that long-term application of compost resulted in a significant ^{15}N -enrichment of soils and plants relative to fertilizer. Therefore, this study suggested that $\delta^{15}\text{N}$ could serve as promising indicators of organic fertilizers application when used with other independent evidence. However, further studies under many conditions should be conducted to prepare reliable $\delta^{15}\text{N}$ guidelines for organic produce, since the $\delta^{15}\text{N}$ of inorganic soil-N and plant-N are influenced by various factors such as soil type, plant species, the rate of N application, and processes such as mineralization, nitrification, and denitrification.

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1. Introduction

In recent years, agriculture has entered a period of major change. With increasing interest in sustaining economically viable crop production with minimal environmental impacts, farming without synthetic fertilizers and pesticides (organic farming) has been widely adopted as an alternative agricultural practice (Reganold et al., 1993; Kirchmann and Thorvaldsson, 2000). Market analyses currently show an increase in consumer demand for organic produce (Rigby et al., 2001). With the increased economic rewards of organic farming, consumer concerns have grown about

whether organically-labeled produce is truly grown with organic inputs.

Conventional and organic farming systems have often been compared to increase the understanding of the organic systems. Such studies have focused on nitrate leaching (Watson et al., 1993), nitrous oxide emission (Li et al., 2002), soil organic matter dynamics (Willson et al., 2001), soil biological properties (Fließbach and Mäder, 2000), and soil physical properties (Hansen, 1996). One powerful tool to study plant–soil N dynamics is measurements of natural ^{15}N composition ($\delta^{15}\text{N}$) (Högberg, 1997). However, few studies have been conducted on $\delta^{15}\text{N}$ patterns of soil and plant between conventional and organic fields, although $\delta^{15}\text{N}$ can be an indicator of N source or cycling patterns (Choi et al., 2002).

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The use of composted manure has increased in some areas where the use of chemical fertilizer is prohibited or organic agriculture is encouraged (Choi et al., 2001a). In general, composted manure has a higher $\delta^{15}\text{N}$ than fertilizer due to faster ammonia volatilization of the lighter N isotope (^{14}N) than ^{15}N during the composting process (Kerley and Jarvis, 1996), with values between -3 and 2‰ for fertilizer (Wassenaar, 1995) and above 8‰ for fresh or composted manure (Wassenaar, 1995; Choi et al., 2002). This $\delta^{15}\text{N}$ difference between N inputs suggests that N isotope signatures of plants and soils could differ between organic and conventional farming systems. Although plant $\delta^{15}\text{N}$ is generally not a conservative tracer of N sources applied to fields since it varies temporally depending on the interactions between N sources and N isotope fractionation (Handley and Scrimgeour, 1997; Robinson, 2001), Choi et al. (2002) observed that compost-treated plants were more enriched in ^{15}N than those receiving fertilizers during early growth. However, few studies have been conducted to examine the $\delta^{15}\text{N}$ pattern of soils and plants from cropping fields subjected to long-term fertilizer or compost treatments (Yoneyama et al., 1990).

The high biological activity of soil inorganic-N (NH_4^+ and NO_3^-) means that levels of inorganic soil-N will change quickly in response to various N transformation processes including nitrification and denitrification. Because many processes of the N cycle fractionate against the heavy isotope (^{15}N), $\delta^{15}\text{N}$ patterns of inorganic soil-N will also be sensitive to N transformation processes (Högberg, 1997). Nitrification is the dominant N process responsible for ^{15}N enrichment of the remaining NH_4^+ in aerobic soil, while denitrification is mainly responsible for ^{15}N enrichment of the remaining NO_3^- in anaerobic soil (Mariotti et al., 1981; Choi et al., 2001b). However, Rice et al. (1988) and Aulakh et al. (2000) observed stimulated denitrification in partially anaerobic region of aerobic soils due to increased microbial demand for O_2 during decomposition of organic substances, and this suggests that the incorporation of organic inputs would increase the $\delta^{15}\text{N}$ of NO_3^- in compost-applied upland soil due to enhanced denitrification.

In this study, the difference in $\delta^{15}\text{N}$ of soil-N (total-, NH_4^+ -, and NO_3^- -N) and plant-N between conventional and organic upland fields was investigated as a possible marker of organic versus chemical fertilizer management. Although rates of denitrification and nitrification were not actually measured, comparing $\delta^{15}\text{N}$ and N concentrations on the NH_4^+ and NO_3^- pools could provide comparable insight into N dynamics of conventional and organic soils.

2. Materials and methods

2.1. Study area

The study area is 30 km east of Seoul, Korea ($127^\circ 18' \sim 22^\circ \text{E}$, $37^\circ 31' \sim 35^\circ \text{N}$) in a temperate climate

zone. Monthly mean temperature ranges from 20 to 26 °C during summer and from -5 to 1 °C during winter. The annual precipitation is about 1400 mm with nearly 70% of total precipitation in summer.

Upland soils in the study area are mainly sandy loam developed from alluvial deposits (Rural Development Administration, 2000). As this area is close to the North Han River, which is the main source of local drinking water, organic farming practices have been recommended since the 1990s to reduce contaminant input into surface water. Thus, fields can be classified into fertilizer-applied and compost-applied fields. The dominant species of plants were *Zea mays* L., *Solanum melongena* L., *Capsicum frutescens* L., *Brassica oleracea* L., *Cucumis sativas* L., *Lactuca sativa* L., *Spinacia oleracea* L., *Perilla ocymoides* L., and *Sesamum indicum* L.

2.2. Samplings

Forty cropping fields, consisting of 20 fertilizer-treated upland fields (FU) and 20 compost-treated upland fields (CU), were selected from the study area. Fertilizers, mainly urea, were applied at rates of 200–400 kg N ha⁻¹ yr⁻¹ in FU fields, while compost was applied at a rate of 300–600 kg N ha⁻¹ yr⁻¹ in CU fields. Basal N dressing was usually applied in the late spring. In CU fields, compost has been applied for at least 5 years according to the governmental regulation of organic farming. Each field has been subjected to slightly different agricultural practices such as crop species and rates of N-inputs.

Soils and plants were sampled in the early summer of 1999, corresponding to the early growing stage of plants in the fields. Plants at this stage probably show $\delta^{15}\text{N}$ differences between fertilizer and compost applications better than plants in the later growing season (Choi et al., 2002). In each field, aboveground parts of five plants were randomly collected and a composite soil sample was made using five sub-samples taken individually from five root zones (0–30 cm in depth). In addition, eight types of fertilizer (urea, ammonium sulfate, and six types of compound fertilizers) and 10 samples of composts used in the study area were collected for $\delta^{15}\text{N}$ analysis.

2.3. Analyses

Plant samples were dried at 60 °C and soil samples were air-dried. Plant and soil samples were ground into very fine powders and used for total-N analysis. Total-N concentrations and the corresponding $\delta^{15}\text{N}$ of plant and soil samples were measured using a continuous-flow stable isotope ratio mass spectrometer (IsoPrime-EA, Micromass, UK) linked with a CN analyzer.

About 100 g of fresh soil was extracted with 250 ml of 2 M KCl to determine the concentrations and $\delta^{15}\text{N}$ of NH_4^+ and NO_3^- in soil samples. Some portions of the extracts were added to distillation flasks and steam-distilled with

MgO for NH_4^+ ; thereafter the samples in flask were distilled again after addition of Devarda's alloy for NO_3^- determination. The liberated NH_3 was collected into H_3BO_3 -indicator solution (Keeney and Nelson, 1982). The N concentrations of NH_4^+ and NO_3^- were determined by titration with standard H_2SO_4 solution. Remaining extracts were used for the analysis of $\delta^{15}\text{N}$ of soil inorganic-N. The NH_4^+ and NO_3^- in the remaining extracts were steam-distilled according to the above method and collected into H_2SO_4 instead of H_3BO_3 -indicator solution. For samples with low inorganic N concentration, two to ten replicated extracts were distilled separately and combined for $\delta^{15}\text{N}$ analysis. To prevent isotopic cross-contamination between samples, 25 ml of reagent-grade ethanol was added to distillation flasks and steam-distilled for 3 min between each sample distillation (Hauck, 1982). After the adjustment of the solution pH to 2–3 using 0.1 N H_2SO_4 , the solution was concentrated under an infra-red lamp. The NH_4^+ in the solution was converted into N_2 gas under vacuum through oxidation using alkaline LiOBr (Hauck, 1982). The N_2 gas was analyzed for $\delta^{15}\text{N}$ on a dual-inlet stable isotope ratio mass spectrometer (Optima, Micro-mass, UK) with triple traps for purifying the gas. Although steam distillation method may cause N isotope fractionation during isolation of N (Robinson, 2001), the accuracy and reproducibility of the analytical procedure checked with reference materials (RM 8548: IAEA-N2 and RM 8549: IAEA-N3) from the International Atomic Energy Agency (Vienna, Austria) were better than 0.4 and 0.2‰, respectively.

Compost samples were air-dried, ground, and analyzed for $\delta^{15}\text{N}$ using the continuous-flow stable isotope ratio mass spectrometer. Fertilizer-N was Kjeldahl-digested (Hauck, 1982), and $\delta^{15}\text{N}$ of N_2 gas was determined through the same procedure described above.

2.4. Calculations and statistical analysis

The N isotopic signature ($\delta^{15}\text{N}$) of a sample was expressed in parts per thousand deviation from the atmospheric N_2 as defined by the following equation:

$$\delta^{15}\text{N} (\text{‰}) = [(R_{\text{sample}}/R_{\text{standard}}) - 1] \times 1000 \quad (1)$$

where, R_{sample} and R_{standard} are the $^{15}\text{N}/(^{14}\text{N} + ^{15}\text{N})$ ratios of the sample and standard, respectively.

Integrated $\delta^{15}\text{N}$ values of inorganic N consisting of two components (NH_4^+ plus NO_3^-) were calculated as follows (Choi et al., 2002):

$$\delta^{15}\text{N}_{(\text{NH}_4+\text{NO}_3)} = (\delta^{15}\text{N}_{\text{NH}_4} \times C_{\text{NH}_4} + \delta^{15}\text{N}_{\text{NO}_3} \times C_{\text{NO}_3}) / (C_{\text{NH}_4} + C_{\text{NO}_3}) \quad (2)$$

where $\delta^{15}\text{N}_{(\text{NH}_4+\text{NO}_3)}$ is the integrated $\delta^{15}\text{N}$ of NH_4^+ plus NO_3^- and $\delta^{15}\text{N}_{\text{NH}_4}$, $\delta^{15}\text{N}_{\text{NO}_3}$, C_{NH_4} , and C_{NO_3} are the $\delta^{15}\text{N}$ values and concentrations of NH_4^+ and NO_3^- , respectively.

Data were analyzed in a one-way ANOVA using the Generalized Linear Models procedure (SAS Institute, 1989). Least significant differences (LSD) were used to compare differences (at the 95% confidence level) between means for selected parameter, such as N concentrations and $\delta^{15}\text{N}$. Correlation coefficients between N concentration and $\delta^{15}\text{N}$ of various N pools were analyzed using Fisher's z -test.

3. Results

3.1. $\delta^{15}\text{N}$ of fertilizers and composts

The isotope composition of composts was significantly ($P < 0.05$) enriched in ^{15}N relative to fertilizers (Table 1). The $\delta^{15}\text{N}$ of fertilizers was close to 0‰, ranging from −3.9‰ for ammonium sulfate to 0.5‰ for a compound fertilizer (N–P₂O₅–K₂O = 15-15-15). Composts had much higher $\delta^{15}\text{N}$ than fertilizers, ranging between 15.4 and 19.4‰.

3.2. N contents and $\delta^{15}\text{N}$ of soils and plants

Contents of total soil-N and inorganic soil-N (NH_4^+ plus NO_3^-) were significantly ($P < 0.05$) greater in CU than in FU fields (Table 2). In FU fields, NO_3^- contributed more than 80% to total inorganic-N concentration; whereas NH_4^+ and NO_3^- contributed similarly to total inorganic soil-N in CU fields. Total-N pools of CU soils were significantly ($P < 0.05$) enriched in ^{15}N by about 2.9‰ relative to FU soils (Table 3). The $\delta^{15}\text{N}$ of NO_3^- and integrated $\delta^{15}\text{N}$ of NH_4^+ plus NO_3^- were also significantly ($P < 0.05$) higher in CU than in FU soils. However, the NH_4^+ pool did not differ significantly in $\delta^{15}\text{N}$ between the soils. In FU soils, the $\delta^{15}\text{N}$ of NH_4^+ was significantly ($P < 0.05$) higher than that of NO_3^- , whereas the $\delta^{15}\text{N}$ of NH_4^+ and NO_3^- for CU soils did not differ significantly.

Although total-N concentrations of plants did not differ significantly between FU and CU fields, the mean $\delta^{15}\text{N}$ of plants were significantly ($P < 0.05$) higher in CU ($14.6 \pm 3.3\text{‰}$) than in FU ($4.1 \pm 1.7\text{‰}$) fields (Table 4). In addition, irrespective of crop species, the $\delta^{15}\text{N}$ of plants was significantly ($P < 0.05$) higher in CU than in FU fields.

Table 1
Nitrogen isotope compositions ($\delta^{15}\text{N}$) of fertilizers and composts used in the study area

Sample	$\delta^{15}\text{N}$ (‰)		Number of samples
	Mean $\pm$ SD	Range	
Fertilizers ^a	-2.6 ± 0.5	−3.9 to 0.5	8
Composts	17.4 ± 1.2	15.4 to 19.4	10

^a The types of fertilizers are urea, ammonium sulfate, and six compound fertilizers (N–P₂O₅–K₂O, 21–17–17, 15–15–15, 11–12–10, 12–8–12, 13–0–13, and 16–0–12).

Table 2

Nitrogen concentrations of total soil-N and inorganic soil-N from fertilizer-applied and compost-applied fields

Fields	Total-N (g N kg ⁻¹)	NH ₄ ⁺ (mg N kg ⁻¹)	NO ₃ ⁻ (mg N kg ⁻¹)	NH ₄ ⁺ + NO ₃ ⁻ (mg N kg ⁻¹)
Fertilizer-applied	1.4 ± 0.2	6.1 ± 1.3	36.7 ± 15.3	42.7 ± 15.6
Compost-applied	2.0 ± 0.7	24.7 ± 21.5	31.1 ± 19.4	55.8 ± 34.8
LSD (<i>P</i> = 0.05)	0.4	10.2	11.8	7.0

Values are the means ± standard deviations of 20 samples from each field.

Table 3

Nitrogen isotope compositions (δ¹⁵N) of total soil-N and inorganic soil-N from fertilizer-applied and compost-applied fields

Fields	Total-N (‰)	NH ₄ ⁺ (‰)	NO ₃ ⁻ (‰)	NH ₄ ⁺ + NO ₃ ⁻ ^a (‰)	LSD (<i>P</i> = 0.05)
Fertilizer-applied	5.9 ± 0.7	9.0 ± 3.0	4.7 ± 1.1	5.2 ± 1.0	1.4
Compost-applied	8.8 ± 2.0	9.0 ± 8.0	11.6 ± 4.5	11.4 ± 4.4	4.7
LSD (<i>P</i> = 0.05)	1.0	3.9	2.1	2.0	

Values are the means ± standard deviations of 20 samples from each field.

^a Integrated means of δ¹⁵N of NH₄⁺ plus NO₃⁻ calculated using Eq. (2).

3.3. Correlation between N concentrations and δ¹⁵N

Nitrogen concentration and the corresponding δ¹⁵N of each soil-N pool (total-N, NH₄⁺, and NO₃⁻) were not significantly correlated for FU fields (Table 5). However, for CU fields, total soil-N (*r* = +0.62, *n* = 20) and NH₄⁺-N (*r* = +0.68, *n* = 20) were positively correlated with their corresponding δ¹⁵N values, whereas soil NO₃⁻-N was not. Total plant-N concentration was negatively correlated with the corresponding

δ¹⁵N for FU fields (*r* = -0.74, *n* = 20), but not for CU fields.

4. Discussion

4.1. δ¹⁵N of soil-N

The significantly higher δ¹⁵N of total soil-N in CU than FU soils (Tables 2 and 3) could be attributed to

Table 4

Nitrogen contents and δ¹⁵N of plant samples from fertilizer-applied and compost-applied fields

Sample ID	Fertilizer-applied			Compost-applied		
	N (g kg ⁻¹)	δ ¹⁵ N (‰)	Species	N (g kg ⁻¹)	δ ¹⁵ N (‰)	Species
1	32.0	4.9	<i>Zea</i>	42.7	14.6	<i>Zea</i>
2	29.5	3.4	<i>Zea</i>	27.6	20.7	<i>Zea</i>
3	32.0	4.8	<i>Solanum</i>	44.5	13.4	<i>Solanum</i>
4	39.0	0.6	<i>Solanum</i>	51.2	15.3	<i>Capsicum</i>
5	19.7	5.3	<i>Capsicum</i>	25.2	13.6	<i>Capsicum</i>
6	27.2	3.7	<i>Capsicum</i>	25.3	16.6	<i>Brassica</i>
7	24.4	5.0	<i>Capsicum</i>	31.8	16.0	<i>Brassica</i>
8	28.4	5.6	<i>Brassica</i>	37.2	10.5	<i>Cucumis</i>
9	36.3	0.3	<i>Brassica</i>	41.4	16.2	<i>Cucumis</i>
10	21.4	5.0	<i>Brassica</i>	23.5	11.8	<i>Cucumis</i>
11	48.0	2.3	<i>Cucumis</i>	24.7	14.8	<i>Cucumis</i>
12	38.2	2.5	<i>Cucumis</i>	36.1	14.3	<i>Lactuca</i>
13	27.2	2.9	<i>Cucumis</i>	29.8	14.2	<i>Lactuca</i>
14	22.2	5.3	<i>Cucumis</i>	12.6	13.3	<i>Lactuca</i>
15	40.5	3.7	<i>Lactuca</i>	49.3	12.1	<i>Lactuca</i>
16	23.7	4.5	<i>Lactuca</i>	16.0	9.6	<i>Spinacia</i>
17	21.9	4.8	<i>Spinacia</i>	25.8	9.3	<i>Spinacia</i>
18	17.7	6.4	<i>Spinacia</i>	39.3	19.9	<i>Perilla</i>
19	25.0	4.5	<i>Perilla</i>	44.2	21.2	<i>Sesamum</i>
20	16.3	5.7	<i>Sesamum</i>	42.9	14.4	<i>Sesamum</i>
Mean	29.0	4.1		33.6	14.6	
SD	8.4	1.7		10.9	3.3	

Table 5
Correlation coefficients between N content and $\delta^{15}\text{N}$ of each N pool

Fields	Nitrogen pools			
	Soil total-N	Soil NH_4^+ -N	Soil NO_3^- -N	Plant total-N
Fertilizer-applied upland	−0.29 ^a	−0.09 ^a	0.35 ^a	−0.74***
Compost-applied upland	0.62**	0.68**	−0.21 ^a	0.30 ^a

** $P < 0.01$, *** $P < 0.001$.

^a Not significant.

the application of compost enriched in ^{15}N relative to fertilizer as shown in Table 1. Since the $\delta^{15}\text{N}$ of total soil-N is unlikely to change over decades (Johannisson and Högberg, 1994), this result indicates long-term application of compost. Assuming that compost-N contributed all of the increase of total soil-N content of CU over those of FU soils, the $\delta^{15}\text{N}$ of total soil-N in CU ($\delta^{15}\text{N}_{\text{CU}}$) soils could be calculated by using a ^{15}N mass balance based on the mean values of total soil-N contents of CU (TN_{CU} , 2.0 g kg^{-1}) and FU (TN_{FU} , 1.4 g kg^{-1}) soils, $\delta^{15}\text{N}$ of total soil-N in FU soils ($\delta^{15}\text{N}_{\text{FU}}$, 5.9‰), and $\delta^{15}\text{N}$ of compost ($\delta^{15}\text{N}_{\text{Compost}}$, 17.4‰) as follows:

$$\delta^{15}\text{N}_{\text{CU}} = [\text{TN}_{\text{FU}} \times \delta^{15}\text{N}_{\text{FU}} + (\text{TN}_{\text{CU}} - \text{TN}_{\text{FU}}) \times \delta^{15}\text{N}_{\text{Compost}}] / \text{TN}_{\text{CU}} \quad (3)$$

The calculated $\delta^{15}\text{N}_{\text{CU}}$ of 9.4‰ was very close to the measured $\delta^{15}\text{N}$ of 8.8‰ for total soil-N in CU soils, showing that the increased $\delta^{15}\text{N}$ of CU soils was primarily due to the addition of compost. A positive correlation ($r = +0.62$) between the $\delta^{15}\text{N}$ and the N concentration of total soil-N in CU soils (Table 5) further supported the application of ^{15}N -enriched compost as the main factor influencing the increased $\delta^{15}\text{N}$ of CU soils versus FU soils.

The $\delta^{15}\text{N}$ of inorganic soil-N showed similar patterns to total soil-N (Table 3) with compost application resulting in a significantly higher $\delta^{15}\text{N}$ of inorganic N (NH_4^+ plus NO_3^-) in CU than in FU soils. Although we did not measure N fluxes, slightly different N transformations between FU and CU soils could be inferred from the relative ^{15}N enrichment of NO_3^- and NH_4^+ . In FU soils, NH_4^+ was significantly enriched in ^{15}N relative to NO_3^- (Table 3) due to the isotopic fractionation associated with nitrification, which increases the $\delta^{15}\text{N}$ of remaining NH_4^+ in soils (Koba et al., 1998; Choi and Ro, 2003). However, a clear isotopic signal of nitrification was not observed for CU soils since the $\delta^{15}\text{N}$ of NH_4^+ and NO_3^- was virtually the same. The lack of a signal probably arose from more complex N transformations in the microbial community in CU than in FU soils. Because NH_4^+ is the product of mineralization as well as the substrate of nitrification, the $\delta^{15}\text{N}$ of NH_4^+ is affected not only by nitrification, but also by mineralization of N from the organic pool. Therefore, any ^{15}N -enrichment of the NH_4^+

pool resulting from nitrification may have been diminished by replenishment of relatively ^{15}N -depleted NH_4^+ from the organic-N pool (Choi et al., 2002). The more abundant organic-N pool of CU soils would result in a larger decrease in the $\delta^{15}\text{N}$ of NH_4^+ compared to FU soils. Increases in the $\delta^{15}\text{N}$ of NO_3^- through denitrification in CU soils would also contribute in part to the lack of a $\delta^{15}\text{N}$ difference between NH_4^+ and NO_3^- . Although CU soils were apparently aerobic, incorporation of organic input would make soil environment partially anaerobic by increasing microbial O_2 demand, and thus enhance the rates of denitrification (Rice et al., 1988; Aulakh et al., 2000).

4.2. Plant $\delta^{15}\text{N}$

Generally, variations in $\delta^{15}\text{N}$ of plants are not simple to interpret since $\delta^{15}\text{N}$ values are affected by various factors, such as the source of N, the depth in soil from which N is taken up, the form of N used (e.g. NH_4^+ , NO_3^- , organic N sources), concentrations of inorganic N, influence of mycorrhizal symbiosis, and fractionations during and after N uptake by plants (Nadelhoffer et al., 1996). Nevertheless, in this study, the consistently higher $\delta^{15}\text{N}$ of plants from CU than those from FU fields (Table 4) could be primarily attributed to the application of isotopically different N inputs. The significantly different $\delta^{15}\text{N}$ of inorganic soil-N (NH_4^+ plus NO_3^-) between CU and FU fields (Table 3) suggests that application of composts increased the $\delta^{15}\text{N}$ of inorganic-N pools, which then led to enrichment of plants in ^{15}N (Choi et al., 2002).

On the other hand, the effect of fertilizer application on the decrease in plant $\delta^{15}\text{N}$ could be inferred from the negative correlation ($r = -0.74$) between the N concentration and the $\delta^{15}\text{N}$ of plants in FU fields (Table 5), since this correlation suggests that plant $\delta^{15}\text{N}$ decreased with increasing contribution of fertilizer-derived N to total plant-N. By using paired natural abundance and tracer ^{15}N fertilizer treatments, Bremer et al. (1993) also concluded that plant $\delta^{15}\text{N}$ at natural abundance levels increased with decreasing contribution of fertilizer-N to total plant-N contents.

Comparing plant $\delta^{15}\text{N}$ with soil $\delta^{15}\text{N}$, plants from CU fields were enriched in ^{15}N , while those from FU fields

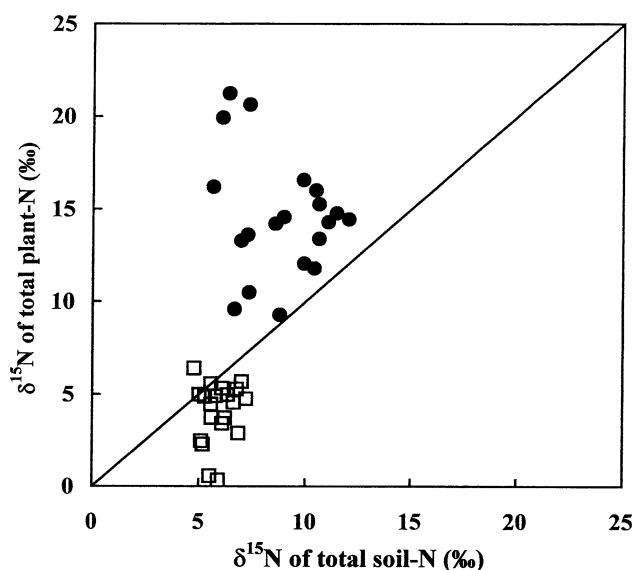


Fig. 1. Crossplot of $\delta^{15}\text{N}$ values of total soil-N versus $\delta^{15}\text{N}$ values of total plant-N from fertilizer-applied ($\square$), compost-applied ($\bullet$) upland fields. The 1:1 line of plant $\delta^{15}\text{N}$: soil $\delta^{15}\text{N}$ is indicated.

were depleted relative to soils (Fig. 1). Since the $\delta^{15}\text{N}$ of N inputs (fertilizer for FU and compost for CU fields) was lower than that of total soil-N for FU but higher for CU soils, these results suggested that $\delta^{15}\text{N}$ of plants was largely affected by N derived from compost or fertilizer rather than indigenous soil-N during the early growth season. This result indicated that the ^{15}N enrichment of

soil relative to plants as well as the absolute $\delta^{15}\text{N}$ values of soil and plants would differ between compost and fertilizer application.

In summary, the application of compost resulted in a significant ^{15}N enrichment of total soil-N, inorganic soil-N (NH_4^+ plus NO_3^-), and total plant-N relative to fertilizer amendment. Although this study lacks supporting information to clearly explain whether the effect of isotopically different N sources on the $\delta^{15}\text{N}$ of inorganic soil-N and plant-N remained significant throughout the entire plant growth, the differences in $\delta^{15}\text{N}$ between compost and fertilizer applications indicate that $\delta^{15}\text{N}$ is a potential marker of organic production of crops. However, since the $\delta^{15}\text{N}$ of soil and plant N pools are sensitive to various factors affecting N dynamics (Handley and Scrimgeour, 1997; Evans, 2001), the effect of growth period, soil type, plant species, the rate and frequency of N application, and N fluxes related to isotope fractionation on the $\delta^{15}\text{N}$ variations need further study.

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Table S1

Nitrogen concentrations and $\delta^{15}\text{N}$ of total soil-N and inorganic soil-N from fertilizer-applied upland fields

Sample ID	Total-N		NH_4^+		NO_3^-		$\text{NH}_4^+ + \text{NO}_3^-$	
	N (g kg^{-1})	$\delta^{15}\text{N}$ (‰)	N (mg kg^{-1})	$\delta^{15}\text{N}$ (‰)	N (mg kg^{-1})	$\delta^{15}\text{N}$ (‰)	N (mg kg^{-1})	$\delta^{15}\text{N}$ (‰) ^a
1	1.6	5.8	5.6	7.5	10.5	3.3	16.1	4.8
2	1.5	6.1	7.1	8.9	55.2	5.7	62.3	6.1
3	1.4	5.3	7.2	8.3	47.3	4.3	54.5	4.8
4	1.5	5.5	7.6	5.1	45.2	5.5	52.8	5.4
5	1.0	6.8	5.6	6.9	70.6	5.4	76.2	5.5
6	1.4	6.2	4.3	9.2	52.1	3.7	56.4	4.1
7	1.5	6.4	5.5	4.9	39.8	5.4	45.3	5.3
8	1.0	5.6	6.2	2.4	13.7	1.8	19.9	2.0
9	1.2	5.9	6.7	10.9	27.1	3.6	33.8	5.0
10	1.4	5.0	4.6	8.7	25.4	4.1	30.0	4.8
11	1.7	5.2	5.6	4.4	36.6	5.7	42.2	5.5
12	1.6	5.1	6.9	5.1	32.7	4.6	39.6	4.7
13	1.4	6.9	6.7	11.1	47.9	4.5	54.6	5.3
14	1.2	6.1	4.9	2.4	24.5	5.7	29.4	5.2
15	1.3	5.6	7.1	7.9	58.2	4.4	65.3	4.8
16	1.5	5.6	4.1	9.3	29.1	6.3	33.2	6.7
17	1.5	7.2	6.4	5.3	32.3	4.5	38.7	4.6
18	1.5	4.8	8.7	8.5	24.1	5.4	32.8	6.2
19	1.7	6.6	3.9	12.9	26.8	4.7	30.7	5.7
20	1.0	7.0	6.5	11.6	35.1	5.8	41.6	6.7
Mean	1.4	5.9	6.1	9.0	36.7	4.7	42.7	5.2
SD	0.2	0.7	1.3	3.0	15.3	1.1	15.6	1.0

^a Integrated means of $\delta^{15}\text{N}$ of NH_4^+ plus NO_3^- calculated using Eq. (2).

Table S2

Nitrogen concentrations and $\delta^{15}\text{N}$ of total soil-N and inorganic soil-N from compost-applied upland fields

Sample ID	Total-N		NH_4^+		NO_3^-		$\text{NH}_4^+ + \text{NO}_3^-$	
	N (g kg ⁻¹)	$\delta^{15}\text{N}$ (‰)	N (mg kg ⁻¹)	$\delta^{15}\text{N}$ (‰)	N (mg kg ⁻¹)	$\delta^{15}\text{N}$ (‰)	N (mg kg ⁻¹)	$\delta^{15}\text{N}$ (‰) ^a
1	1.7	9.0	8.6	1.9	35.2	9.0	43.8	7.6
2	1.2	7.3	13.3	5.7	23.2	11.2	36.5	9.2
3	2.2	10.6	23.1	2.4	16.6	20.0	39.7	9.8
4	2.6	10.6	12.1	6.9	78.6	6.2	90.7	6.3
5	1.9	7.3	6.3	2.3	25.0	14.9	31.3	12.4
6	2.7	9.9	59.4	19.8	15.5	7.3	74.9	17.2
7	3.4	10.5	20.8	11.6	20.5	15.3	41.3	13.4
8	1.6	7.3	60.1	8.8	51.6	10.8	111.7	9.7
9	1.4	5.6	11.9	0.3	10.2	6.7	22.1	3.2
10	2.1	10.4	17.9	6.3	16.9	5.5	34.8	5.9
11	3.5	11.4	78.7	21.6	80.7	9.4	159.4	15.4
12	1.6	11.1	15.1	7.9	21.4	13.6	36.5	11.2
13	1.7	8.6	19.6	9.5	20.3	16.8	39.9	13.2
14	1.3	7.0	10.4	1.3	28.2	7.5	38.6	5.8
15	1.7	9.9	18.7	10.6	40.0	13.1	58.7	12.3
16	2.1	6.7	18.2	27.6	17.5	9.6	35.7	18.8
17	2.5	8.8	24.6	9.9	25.4	12.5	50.0	11.2
18	1.3	6.1	6.6	2.2	22.0	11.6	28.6	9.4
19	1.1	6.4	7.4	0.9	33.5	21.6	40.9	17.9
20	1.8	12.0	60.5	22.5	40.6	9.3	101.1	17.2
Mean	2.0	8.8	24.7	9.0	31.1	11.6	55.8	11.4
SD	0.7	2.0	21.5	8.0	19.4	4.5	34.8	4.4

^a Integrated means of $\delta^{15}\text{N}$ of NH_4^+ plus NO_3^- calculated using Eq. (2).

Appendix A

Supplementary information 1

See Table S1

Appendix B

Supplementary information 2

See Table S2

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